

AMD discontinues endorsement of SysMark

AMD discontinued the endorsement of SysMark 2012 Benchmark since it claims their results are biased

Pentax mini lens

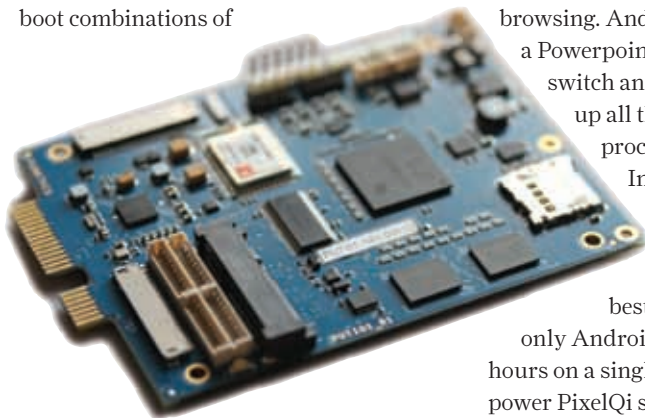
Pentax introduces small size interchangeable lens

think of keeping your notebook at home on an important business trip? Nope. This brings us to the scenario we painted for you at the beginning.

What if we could use low power and low performance systems for specific tasks on the same machine? Do you really need a Core i7 when you're just checking mails? The answer is no, but x86 based CPUs cannot throttle down so drastically. Hence the need for hybrid devices.

Enter Cupp Computing with its PunkThis module

At Computex we came across a non-descript booth with a MacBook and Asus Netbook on which something was being demoed. Curiosity got the better of us when we noticed the MacBook was running Android. Now, Android on Macbooks isn't a big deal and neither is dual boot combinations of



PT Module

various kinds. What caught our eye was that at the demonstrator's beck and call the OS was instantly switching between Mac OS X and Android!

Hybrid or Multi-Mode Computing allows scalable and task specific computing; that is, the correct processing capacity for the task. This allows much greater functionality to be integrated into a standard PC architecture. Security, Always On and Context Awareness are growing trends with Computing and difficult to accommodate on a standard PC. The other side of the coin is that today's smart devices (phones and tablets) lack the utility of the general purpose processor that the PC provides. Thus making productivity difficult with these devices. "The future lies in the closer integration

of platforms. As we develop our platform we will weave these platforms into a more seamless "single environment," said Omar Nathaniel, President – CTO, CUPP Computing.

This switchable scenario is made possible by an SoC module having an ARM processor that fits into your x86 platform based netbook's 2.5-inch SATA hard drive bay. In the module, you get a 1GHz TI OMAP DM3730 ARM processor paired with 512MB of RAM, a Mini PCIe SSD slot to replace the host storage and shared flash memory for both systems.

In essence, PunkThis offers the ability to combine a low power consuming ARM processor with a full-fledged PC experience offered by a high performance Intel x86 processor. The idea is simple: switch instantaneously to the Android OS when performing simple tasks such as browsing. And should you need to make a Powerpoint presentation, flick a switch and you're in Windows using up all the precious wattage and processing power you require.

In a mixed usage scenario, a Punked system can run on Android for about 20 hours at a stretch. In the best case scenario, it can run only Android for a mind-boggling 40 hours on a single charge (with another low power PixelQi screen mod). As of now, at a price of around \$200, it's certainly steep for a mod. But as Nathaniel said, CUPP would be releasing a reference platform in December. Further to which we suspect economies of scale that come with mass manufacturing may bring this price point down considerably.

A reference platform that is essentially open, will result in some interesting designs and customisations from OEM and ODM partners. Besides this, the SoC module as part of a desktop enclosure has its uses for developers to test apps on. As a stand-alone computer it can be used as media centre, terminal or connected device. When used in conjunction with a desktop, PunkThis provides an ideal instant on, low power system to check mail and surf the web.

Developers gain the ability to target the ARM v7 code base on physical hardware while retaining the functionality of a desktop environment.

What's next?

All of this sounds promising but could there be game changers in the offing that could make even this hybrid format redundant? The x86 v/s ARM platform war has seen a lot of twists and turns. Intel keeps making announcements about better low-power consumption processors for mobiles but analysts believe they still have a long way to go. Meanwhile, ARM with its eagle core Cortex A15 is talking of quad core 2.5 GHz speeds, 20 nm manufacturing, virtualisation, and server-scale computing. We see the two warring factions making inroads into each other's territories here.

Also, because of the close association between x86, Intel and Microsoft, even the platform is popularly called Wintel. The latest twist came at CES 2011 when Microsoft announced that the next generation of Windows will be compatible with ARM chips. With this announcement, x86 loses its biggest ally and the Wintel banner may finally fall.

The new version of the world's favourite OS will work with natively-compiled ARM code and would require some work with respect to drivers, controllers, IO buses etc. But proof of concept was seen. Could Windows 8 then be the OS of choice for today's landscape where

demands on computing systems is high and portability, power efficiency is a must?

With Eagle core chips (or those even further down the ARM roadmap) we could have

full computing devices that are Windows based, powerful, run cool and the best part; have battery life in the region of 40 plus hours. Also, if it's Windows playing ring-master the rest of the players in the ecosystem – application and hardware peripheral makers – will easily toe the line and dish out software and drivers for the instruction set very quickly.

Whatever the outcome, it'll be interesting to watch how events unfold. **d**



Arm Cortex a15